

APPENDIX 7 – DETAILED OUTCOME CATEGORY SUMMARY

Outcome Group: Democratic Practices

DP1: Decentralized Organizational Governance

Group: Democratic Practices

Top Conditions Present:

- GV1: Decentralized Authority Distribution (Number of cases = 6)
- RS3: Economic Value Redistribution (3)
- RS1: Community Capacity Development (2)
- TP4: Community-Centered Design Processes (2)
- TP2: Open Development and Transparency (2)
- TT1: Community-Controlled Physical Infrastructure (2)

Unique Case Count: 10 cases

Cases: CoopCycle, DAIR (Distributed AI Research Institute), Detroit Community Technology Project (CTC), Enspiral, Guifi.net, Loomio, Nea Guinea/Tzoumakers, Participatory Brazil, Sensorica, The Drivers Cooperative

Conditions combine as follows to produce this outcome: GV1 (Decentralized Authority Distribution) *structures authority* by configuring non-hierarchical decision-making architectures, while RS3 (Economic Value Redistribution) *incentivizes participation* through material stakes in collective governance. RS1 (Community Capacity Development) *builds agency* by developing the skills needed to exercise distributed authority, and TP4 (Community-Centered Design Processes) *facilitates deliberation* through participatory design that centers affected communities. TP2 (Open Development and Transparency) *enables participation* by creating scrutiny mechanisms, while TT1 (Community-Controlled Physical Infrastructure) *operationalizes practice* by grounding governance in material assets communities actually control. Together, these conditions create the structural, material, and procedural foundations for collective self-governance—GV1 provides the governance architecture, RS3 and TT1 provide material foundations that create stakes and operational capacity, while RS1, TP2, and TP4 provide the participatory processes through which authority is exercised.

DP2: Community Data Sovereignty and Control

Group: Democratic Practices

Top Conditions Present:

- TP1: Democratic Data Authority (6)
- PO1: Policy and Systemic Transformation Advocacy (3)
- TP2: Open Development and Transparency (2)

- TT3: Indigenous Data Sovereignty Infrastructure (2)
- GV3: Federated Multi-Scale Coordination (2)
- PU1: Indigenous Epistemological Centering (2)

Case Count: 8 cases

Cases: DECODE, First Nations Information Governance Centre (FNIGC), Global Indigenous Data Alliance (GIDA), MAIAM NAYRI WINGARA, Matrix, MiData, OpenSAFELY, Worker Algorithm Observatory (WAO)

Conditions combine as follows to produce this outcome: TP1 (Democratic Data Authority) *establishes control* by creating governance mechanisms where data subjects have authority over how their data is managed, while PO1 (Policy and Systemic Transformation Advocacy) *legitimizes sovereignty* by securing external recognition and policy frameworks that validate community data governance. TP2 (Open Development and Transparency) *enables scrutiny* through open processes that counter proprietary enclosure, and TT3 (Indigenous Data Sovereignty Infrastructure) *instantiates sovereignty* by building purpose-built systems for Indigenous data governance. GV3 (Federated Multi-Scale Coordination) *federates control* by enabling data governance across multiple scales while preserving local authority, and PU1 (Indigenous Epistemological Centering) *grounds authority* by anchoring data governance in Indigenous worldviews rather than Western frameworks. Together, these conditions construct data sovereignty through technical, political, and epistemological dimensions—TP1 and TT3 provide the technical governance infrastructure, PO1 secures external legitimacy, while GV3 enables scalar coordination and PU1 ensures epistemological grounding.

DP3: Community-Controlled Infrastructure

Group: Democratic Practices

Top Conditions Present:

- TT1: Community-Controlled Physical Infrastructure (5)
- RS2: Access and Skills Democratization (2)
- TP3: Multi-Scale Architecture Design (2)
- OI3: Anti-Extractive Value Circulation (2)

Case Count: 5 cases

Cases: Detroit Community Technology Project (CTC), Guifi.net, SOLshare, Telecomunicaciones Indígenas Comunitarias (TIC A.C.), Zenzeleni

Conditions combine as follows to produce this outcome: TT1 (Community-Controlled Physical Infrastructure) *materializes control* by creating community-owned physical networks and systems, while RS2 (Access and Skills Democratization) *skills operation* by building local capacity to maintain and govern infrastructure. TP3 (Multi-Scale Architecture Design) *designs scale* by creating technical architectures that enable both local control and broader coordination, and OI3 (Anti-Extractive Value Circulation) *defines anti-extraction* by establishing organizational forms that prevent external value capture. Together, these

conditions construct infrastructure control through material ownership combined with operational capacity—TT1 provides the physical foundation, RS2 ensures communities can actually operate what they own, TP3 enables coordination without sacrificing local autonomy, and OI3 ensures value remains in community hands.

DP4: Inclusive Participation and Representation

Group: Democratic Practices

Top Conditions Present:

- SC1: Inclusive Participation Structures (4)
- RS1: Community Capacity Development (3)
- RS2: Access and Skills Democratization (2)
- PO2: Resistance-Opposition Integration (2)
- OI5: Local-Global Scalar Positioning (2)
- PT4: Expertise-Community Knowledge Integration (2)

Case Count: 6 cases

Cases: CLEAR (Civic Laboratory for Environmental Action Research), Digital Empowerment Foundation, Les Mercedes, Masakhane, Participatory Brazil, Stop LAPD Spying Coalition

Conditions combine as follows to produce this outcome: SC1 (Inclusive Participation Structures) *enables inclusion* by structuring participation to reach marginalized groups, while RS1 (Community Capacity Development) *capacitates participation* by building the skills and resources communities need to engage effectively. RS2 (Access and Skills Democratization) *democratizes skills* by removing barriers to participation, and PO2 (Resistance-Opposition Integration) *challenges exclusion* by combining opposition to exclusionary systems with constructive alternatives. OI5 (Local-Global Scalar Positioning) *positions globally* by enabling local voices to connect to transnational networks, and PT4 (Expertise-Community Knowledge Integration) *integrates knowledge* by bridging technical expertise with community knowledge. Together, these conditions address both structural barriers to participation and capacity gaps—SC1 and PO2 address structural exclusion, RS1 and RS2 build participatory capacity, while OI5 and PT4 connect local participation to broader networks and expertise.

DP5: Open Access and Transparent Systems

Group: Democratic Practices

Top Conditions Present:

- TP2: Open Development and Transparency (4)
- TT4: Participatory Design Platforms (3)

- PO1: Policy and Systemic Transformation Advocacy (3)
- GV1: Decentralized Authority Distribution (2)
- TP3: Multi-Scale Architecture Design (2)

Case Count: 7 cases

Cases: Linux Foundation, Loomio, Matrix, Open Mobility Foundation, OpenStreetMap, Participatory Brazil, The Drivers Cooperative

Conditions combine as follows to produce this outcome: TP2 (Open Development and Transparency) *creates transparency* by establishing open processes that enable scrutiny and modification, while TT4 (Participatory Design Platforms) *operationalizes transparency* through technical systems enabling collective participation. PO1 (Policy and Systemic Transformation Advocacy) *advocates openness* by securing policy frameworks that support open access, and GV1 (Decentralized Authority Distribution) *distributes access* by preventing concentrated control over systems and resources. TP3 (Multi-Scale Architecture Design) *scales access* by designing systems that work across multiple contexts while maintaining openness. Together, these conditions produce openness through technical design, governance structure, and policy advocacy—TP2 and TT4 provide the technical mechanisms for transparency, GV1 ensures distributed rather than concentrated control, TP3 enables scaled access, and PO1 secures external recognition for open approaches.

DP6: Indigenous and Cultural Governance Integration

Group: Democratic Practices

Top Conditions Present:

- PU1: Indigenous Epistemological Centering (4)
- PO1: Policy and Systemic Transformation Advocacy (3)
- PT2: Multi-Network Strategic Positioning (2)
- TT3: Indigenous Data Sovereignty Infrastructure (2)

Case Count: 4 cases

Cases: First Nations Information Governance Centre (FNIGC), Global Indigenous Data Alliance (GIDA), Te Hiku Media, Te Mana Raraunga

Conditions combine as follows to produce this outcome: PU1 (Indigenous Epistemological Centering) *grounds governance* by anchoring governance structures in Indigenous worldviews and knowledge systems rather than Western frameworks, while PO1 (Policy and Systemic Transformation Advocacy) *legitimizes frameworks* by securing external recognition for Indigenous governance approaches. PT2 (Multi-Network Strategic Positioning) *builds solidarity* by connecting Indigenous initiatives across networks for collective advocacy, and TT3 (Indigenous Data Sovereignty Infrastructure) *instantiates governance* by building technical systems that embody Indigenous data governance principles. Together, these conditions construct cultural governance integration through epistemological grounding, technical instantiation, network solidarity, and policy legitimization—PU1 provides the

foundational orientation without which governance defaults to Western frameworks, TT3 translates principles into technical reality, PT2 builds collective power, and PO1 secures external validation.

DP7: Legal and Institutional Protection

Group: Democratic Practices

Top Conditions Present:

- RG1: Legal Innovation for Sovereignty Protection (4)
- RS3: Economic Value Redistribution (3)
- TT1: Community-Controlled Physical Infrastructure (2)
- PU1: Indigenous Epistemological Centering (2)
- GV4: Democratic Scaling Strategies (2)
- RG3: Strategic Regulatory Navigation (2)

Case Count: 6 cases

Cases: CoopCycle, First Nations Information Governance Centre (FNIGC), Guifi.net, Solar Commons, Te Hiku Media, Telecomunicaciones Indígenas Comunitarias (TIC A.C.)

Conditions combine as follows to produce this outcome: RG1 (Legal Innovation for Sovereignty Protection) *formalizes protection* by creating novel legal frameworks that secure community sovereignty, while RS3 (Economic Value Redistribution) *stakes protection* by creating material interests in maintaining protective structures. TT1 (Community-Controlled Physical Infrastructure) *anchors legally* by providing material assets around which legal protections can be constructed, and PU1 (Indigenous Epistemological Centering) *grounds rights* by establishing the epistemological basis for sovereignty claims. GV4 (Democratic Scaling Strategies) *institutionalizes* by creating governance structures that can scale while maintaining protection, and RG3 (Strategic Regulatory Navigation) *navigates regulation* by engaging existing frameworks strategically. Together, these conditions produce legal protection through innovative legal frameworks combined with material anchoring and epistemological grounding—RG1 creates new legal forms, TT1 and RS3 provide material foundations, PU1 establishes rights claims, and GV4 and RG3 enable strategic engagement with institutional contexts.

DP8: Resistance and Accountability Mechanisms

Group: Democratic Practices

Top Conditions Present:

- RS1: Community Capacity Development (2)
- TT6: Algorithmic Transparency Infrastructure (2)
- PO1: Policy and Systemic Transformation Advocacy (2)

- SC1: Inclusive Participation Structures (2)
- PO2: Resistance-Opposition Integration (2)

Case Count: 4 cases

Cases: CLEAR (Civic Laboratory for Environmental Action Research), DAIR (Distributed AI Research Institute), Stop LAPD Spying Coalition, Worker Algorithm Observatory (WAO)

Conditions combine as follows to produce this outcome: RS1 (Community Capacity Development) *enables resistance* by building community capacity to challenge harmful systems, while TT6 (Algorithmic Transparency Infrastructure) *exposes systems* by creating technical tools that make opaque algorithmic processes visible. PO1 (Policy and Systemic Transformation Advocacy) *challenges systems* by engaging policy arenas to demand accountability, and SC1 (Inclusive Participation Structures) *structures resistance* by ensuring those most affected lead accountability efforts. PO2 (Resistance-Opposition Integration) *integrates opposition* by combining resistance with constructive alternatives. Together, these conditions produce accountability through community capacity, technical exposure, inclusive leadership, and strategic advocacy—RS1 builds the power to resist, TT6 provides evidence for accountability claims, SC1 ensures legitimacy through affected leadership, and PO1 and PO2 translate resistance into systemic change.

DP9: Local Capacity and Self-Governance

Group: Democratic Practices

Top Conditions Present:

- RS2: Access and Skills Democratization (2)
- TT1: Community-Controlled Physical Infrastructure (2)
- GV1: Decentralized Authority Distribution (2)

Case Count: 3 cases

Cases: Detroit Community Technology Project (CTC), Participatory Brazil, Zenzeleni

Conditions combine as follows to produce this outcome: RS2 (Access and Skills Democratization) *enables self-governance* by providing communities with skills and access needed for autonomous operation, while TT1 (Community-Controlled Physical Infrastructure) *grounds locally* by creating material foundations for local governance. GV1 (Decentralized Authority Distribution) *distributes authority* by structuring decision-making at local levels. Together, these conditions produce local self-governance through skills development, material infrastructure, and governance architecture—RS2 ensures communities have capacity, TT1 provides the material substrate for local control, and GV1 structures authority at the appropriate scale.

Outcome Group: Community Value

CV1: Worker and Member Economic Benefits

Group: Community Value

Top Conditions Present:

- GV2: Ownership-Based Democratic Control (5)
- RS3: Economic Value Redistribution (4)
- PU5: Ecological and Social Transformation (2)
- OI3: Anti-Extractive Value Circulation (2)
- OI1: Worker/Community Ownership Models (2)

Case Count: 9 cases

Cases: CoopCycle, Fairbnb.coop, Groupmuse, Les Mercedes, MiData, Namma Yatri, Som Energia, The Drivers Cooperative, Zenzeleni

Conditions combine as follows to produce this outcome: GV2 (Ownership-Based Democratic Control) *retains value* by anchoring economic benefits in member/worker ownership structures, while RS3 (Economic Value Redistribution) *redistributes value* through mechanisms that direct economic returns to workers and communities rather than external shareholders. PU5 (Ecological and Social Transformation) *aligns ecology* by integrating economic benefits with broader social and environmental values, and OI3 (Anti-Extractive Value Circulation) *circulates value* by establishing organizational forms that prevent external extraction. OI1 (Worker/Community Ownership Models) *defines ownership* by establishing the legal and organizational basis for worker/member claims on economic value. Together, these conditions produce economic benefits through ownership structures that anchor value locally, redistribution mechanisms that direct flows to members, and organizational forms that prevent extraction—GV2 and OI1 establish ownership claims, RS3 operationalizes redistribution, OI3 prevents leakage, and PU5 ensures alignment with broader values.

CV2: Open Knowledge and Technology Commons

Group: Community Value

Top Conditions Present:

- TP2: Open Development and Transparency (7)
- TT5: Digital Commons Resources (3)
- PT1: Reciprocal Knowledge Exchange (3)
- OI4: Commons-Based Production Models (3)
- TP6: Commons Resource Stewardship (2)
- PO1: Policy and Systemic Transformation Advocacy (2)

Case Count: 9 cases

Cases: DECODE, Linux Foundation, Loomio, Matrix, OpenStreetMap, P2P Foundation, Research Data Alliance, Sensorica, WikiHouse

Conditions combine as follows to produce this outcome: TP2 (Open Development and Transparency) *constructs commons* through open development processes that create collectively accessible knowledge resources, while TT5 (Digital Commons Resources) *instantiates commons* by producing tangible digital artifacts that embody open access principles. PT1 (Reciprocal Knowledge Exchange) *circulates knowledge* by creating bidirectional flows rather than extractive knowledge relationships, and OI4 (Commons-Based Production Models) *defines production* by establishing organizational forms oriented toward commons rather than proprietary outputs. TP6 (Commons Resource Stewardship) *stewards commons* by creating ongoing governance of shared resources, and PO1 (Policy and Systemic Transformation Advocacy) *advocates commons* by securing policy recognition for commons-based approaches. Together, these conditions produce knowledge commons through open production, instantiated artifacts, reciprocal circulation, and stewardship governance—TP2 and OI4 establish the production orientation, TT5 creates the artifacts, PT1 ensures circulation, TP6 provides ongoing stewardship, and PO1 secures external support.

CV3: Community Infrastructure and Capacity Assets

Group: Community Value

Top Conditions Present:

- TP3: Multi-Scale Architecture Design (3)
- TT1: Community-Controlled Physical Infrastructure (3)
- RS2: Access and Skills Democratization (2)
- RS3: Economic Value Redistribution (2)
- PT1: Reciprocal Knowledge Exchange (2)
- RG1: Legal Innovation for Sovereignty Protection (2)

Case Count: 6 cases

Cases: Australian Research Data Commons, Detroit Community Technology Project (CTC), Digital Empowerment Foundation, Guifi.net, SOLshare, Solar Commons

Conditions combine as follows to produce this outcome: TP3 (Multi-Scale Architecture Design) *designs assets* by creating technical infrastructure that operates effectively across scales while remaining community-controlled, while TT1 (Community-Controlled Physical Infrastructure) *produces assets* by creating material infrastructure that communities own and operate. RS2 (Access and Skills Democratization) *builds skills* by developing local capacity to maintain and govern assets, and RS3 (Economic Value Redistribution) *retains value* by ensuring infrastructure generates returns for communities. PT1 (Reciprocal Knowledge Exchange) *exchanges resources* by connecting communities for mutual learning about asset development, and RG1 (Legal Innovation for Sovereignty Protection) *secures assets* by creating legal frameworks protecting community ownership. Together, these conditions produce infrastructure assets through technical design, material production, capacity development, and legal protection—TP3 and TT1 create the assets, RS2 ensures operational

capacity, RS3 retains generated value, PT1 enables learning, and RG1 provides legal security.

CV4: Culturally-Specific Technology and Knowledge

Group: Community Value

Top Conditions Present:

- TT2: Culturally-Grounded Technology Artifacts (3)
- PU2: Cultural Preservation and Linguistic Sovereignty (3)
- PT3: Community Sovereignty Protection in Partnerships (2)
- PT1: Reciprocal Knowledge Exchange (2)
- PU3: Value System Reconceptualization (2)
- PR3: Cultural Erasure and Heritage Threat (2)

Case Count: 6 cases

Cases: Digital Empowerment Foundation, Groupmuse, MAIAM NAYRI WINGARA, Masakhane, Monlam AI, Te Hiku Media

Conditions combine as follows to produce this outcome: TT2 (Culturally-Grounded Technology Artifacts) *embeds culture* by creating technology that reflects and serves specific cultural contexts, while PU2 (Cultural Preservation and Linguistic Sovereignty) *directs preservation* by orienting technology development toward cultural and linguistic maintenance. PT3 (Community Sovereignty Protection in Partnerships) *protects culture* by ensuring external partnerships don't compromise cultural integrity, and PT1 (Reciprocal Knowledge Exchange) *exchanges knowledge* by creating bidirectional flows that respect cultural protocols. PU3 (Value System Reconceptualization) *reconceptualizes value* by reframing technology value in cultural rather than market terms, and PR3 (Cultural Erasure and Heritage Threat) *preserves heritage* by motivating technology development that counters cultural loss. Together, these conditions produce culturally-specific technology through purposeful cultural embedding, protective partnerships, and resistance to cultural erasure—TT2 creates the artifacts, PU2 and PU3 orient development, PT1 and PT3 structure external relationships, and PR3 provides urgency.

CV5: Alternative Economic Models and Structures

Group: Community Value

Top Conditions Present:

- RS4: Alternative Economic Experimentation (2)
- GV5: Inclusive Expertise Integration (2)

Case Count: 4 cases

Cases: CLEAR (Civic Laboratory for Environmental Action Research), Enspiral, Sensorica, Worker Algorithm Observatory (WAO)

Conditions combine as follows to produce this outcome: RS4 (Alternative Economic Experimentation) *experiments models* by testing new forms of economic organization, while GV5 (Inclusive Expertise Integration) *integrates expertise* by democratizing economic knowledge and decision-making. Together, these conditions produce alternative economic models through deliberate experimentation combined with inclusive expertise—RS4 creates the space for innovation, while GV5 ensures that economic alternatives draw on diverse knowledge rather than replicating expert-dominated conventional models.

Outcome Group: Balance

BL1: Federated Multi-Scale Governance

Group: Balance

Top Conditions Present:

- TP3: Multi-Scale Architecture Design (5)
- GV3: Federated Multi-Scale Coordination (3)
- TT1: Community-Controlled Physical Infrastructure (2)
- PO1: Policy and Systemic Transformation Advocacy (2)

Case Count: 6 cases

Cases: Guifi.net, Matrix, MiData, Open Mobility Foundation, OpenAgriNet, SOLshare

Conditions combine as follows to produce this outcome: TP3 (Multi-Scale Architecture Design) *architects federation* by creating technical systems designed to operate across multiple scales simultaneously, while GV3 (Federated Multi-Scale Coordination) *configures federation* by establishing governance structures that coordinate across scales while preserving local autonomy. TT1 (Community-Controlled Physical Infrastructure) *grounds locally* by providing material foundations for local nodes within federated structures, and PO1 (Policy and Systemic Transformation Advocacy) *advocates coordination* by securing policy frameworks that support federated approaches. Together, these conditions produce federated governance through deliberate scalar design in both technical and governance dimensions—TP3 creates the technical architecture, GV3 structures governance coordination, TT1 anchors local nodes, and PO1 builds external support for federation as a model.

BL2: Multi-Network Coordination and Solidarity

Group: Balance

Top Conditions Present:

- PT2: Multi-Network Strategic Positioning (6)
- PO1: Policy and Systemic Transformation Advocacy (4)
- PU1: Indigenous Epistemological Centering (2)

Case Count: 6 cases

Cases: Enspiral, Global Indigenous Data Alliance (GIDA), Open Mobility Foundation, P2P Foundation, Slum Dwellers International, Te Mana Raraunga

Conditions combine as follows to produce this outcome: PT2 (Multi-Network Strategic Positioning) *enables coordination* by strategically positioning initiatives across multiple networks for mutual support and amplified voice, while PO1 (Policy and Systemic

Transformation Advocacy) *amplifies voice* by converting network positioning into policy influence. PU1 (Indigenous Epistemological Centering) *unites solidarity* by creating shared epistemological foundations for network coordination among Indigenous initiatives. Together, these conditions produce multi-network coordination through strategic positioning, policy amplification, and epistemological solidarity—PT2 establishes the network positions, PO1 converts positioning into influence, and PU1 provides shared foundations for Indigenous solidarity networks.

BL3: Bidirectional Knowledge and Resource Exchange

Group: Balance

Top Conditions Present:

- PT1: Reciprocal Knowledge Exchange (6)
- TP6: Commons Resource Stewardship (2)
- RS6: Values-Aligned Resource Allocation (2)
- PT4: Expertise-Community Knowledge Integration (2)
- TP1: Democratic Data Authority (2)
- RG2: Ethical Governance Framework Development (2)

Case Count: 7 cases

Cases: Australian Research Data Commons, DECODE, Digital Empowerment Foundation, Masakhane, OpenSAFELY, P2P Foundation, Research Data Alliance

Conditions combine as follows to produce this outcome: PT1 (Reciprocal Knowledge Exchange) *structures exchange* by creating formal mechanisms for bidirectional rather than extractive knowledge flows, while TP6 (Commons Resource Stewardship) *stewards exchange* by governing shared resources produced through exchange. RS6 (Values-Aligned Resource Allocation) *aligns exchange* by ensuring resource flows reflect community values rather than external metrics, and PT4 (Expertise-Community Knowledge Integration) *bridges expertise* by creating productive integration between technical expertise and community knowledge. TP1 (Democratic Data Authority) *shares data* by enabling data exchange while maintaining community authority, and RG2 (Ethical Governance Framework Development) *governs exchange* by creating ethical frameworks for equitable exchange relationships. Together, these conditions produce bidirectional exchange through structured reciprocity, values-aligned allocation, and ethical governance—PT1 establishes the exchange structure, PT4 bridges knowledge domains, RS6 and RG2 ensure values alignment, and TP1 and TP6 manage shared resources.

BL4: Community Sovereignty Protection in External Engagement

Group: Balance

Top Conditions Present:

- TT2: Culturally-Grounded Technology Artifacts (3)
- PT3: Community Sovereignty Protection in Partnerships (3)
- OI5: Local-Global Scalar Positioning (2)
- RG3: Strategic Regulatory Navigation (2)
- PU2: Cultural Preservation and Linguistic Sovereignty (2)
- PT2: Multi-Network Strategic Positioning (2)

Case Count: 9 cases

Cases: Fairbnb.coop, Les Mercedes, Linux Foundation, Monlam AI, Nea Guinea/Tzoumakers, Open Mobility Foundation, Slum Dwellers International, Te Hiku Media, Telecomunicaciones Indígenas Comunitarias (TIC A.C.)

Conditions combine as follows to produce this outcome: TT2 (Culturally-Grounded Technology Artifacts) *anchors identity* by creating cultural markers that maintain community distinctiveness in external engagement, while PT3 (Community Sovereignty Protection in Partnerships) *protects sovereignty* by establishing partnership terms that preserve community autonomy. OI5 (Local-Global Scalar Positioning) *positions scale* by strategically locating initiatives to benefit from global connections without surrendering local control, and RG3 (Strategic Regulatory Navigation) *navigates external* by engaging regulatory environments strategically. PU2 (Cultural Preservation and Linguistic Sovereignty) *anchors engagement* by grounding external relationships in cultural integrity, and PT2 (Multi-Network Strategic Positioning) *positions strategically* by selecting network positions that enhance rather than compromise sovereignty. Together, these conditions produce sovereignty protection through cultural anchoring, protective partnerships, strategic positioning, and regulatory navigation—TT2 and PU2 anchor identity, PT2 and PT3 structure external relationships, OI5 positions scale, and RG3 navigates regulatory contexts.

BL5: Global-Local Standards Navigation

Group: Balance

Top Conditions Present:

- TP2: Open Development and Transparency (2)
- RG2: Ethical Governance Framework Development (2)

Case Count: 4 cases

Cases: Australian Research Data Commons, OpenSAFELY, OpenStreetMap, Sensorica

Conditions combine as follows to produce this outcome: TP2 (Open Development and Transparency) *adapts standards* by enabling local adaptation of global standards through transparent processes, while RG2 (Ethical Governance Framework Development) *frames standards* by creating ethical frameworks that bridge global standards and local values. Together, these conditions produce standards navigation through transparent adaptation and ethical framing—TP2 enables the technical processes of adaptation, while RG2 provides the ethical frameworks for navigating between global requirements and local needs.

Outcome Group: Impact

IM1: Policy and Regulatory Change

Group: Impact

Top Conditions Present:

- PO1: Policy and Systemic Transformation Advocacy (9)
- PT2: Multi-Network Strategic Positioning (4)
- RG3: Strategic Regulatory Navigation (3)
- PU1: Indigenous Epistemological Centering (3)
- SC1: Inclusive Participation Structures (2)
- RS1: Community Capacity Development (2)

Case Count: 10 cases

Cases: CLEAR (Civic Laboratory for Environmental Action Research), First Nations Information Governance Centre (FNIGC), Global Indigenous Data Alliance (GIDA), Matrix, Open Mobility Foundation, P2P Foundation, Stop LAPD Spying Coalition, Te Mana Raraunga, Telecomunicaciones Indígenas Comunitarias (TIC A.C.), The Drivers Cooperative

Conditions combine as follows to produce this outcome: PO1 (Policy and Systemic Transformation Advocacy) *directly produces* policy change through sustained advocacy and engagement with policy systems, while PT2 (Multi-Network Strategic Positioning) *amplifies influence* by connecting advocacy across multiple networks for greater impact. RG3 (Strategic Regulatory Navigation) *navigates regulation* by engaging existing regulatory frameworks strategically, and PU1 (Indigenous Epistemological Centering) *grounds advocacy* by establishing the epistemological basis for Indigenous policy claims. SC1 (Inclusive Participation Structures) *enables voice* by structuring participation so affected communities lead policy engagement, and RS1 (Community Capacity Development) *builds advocacy capacity* by developing community skills for sustained policy work. Together, these conditions produce policy change through direct advocacy, network amplification, strategic navigation, and inclusive capacity building—PO1 is the primary mechanism, PT2 and RS1 provide amplification and capacity, RG3 enables strategic engagement, PU1 grounds Indigenous claims, and SC1 ensures affected community leadership.

IM2: Community Capacity Building

Group: Impact

Top Conditions Present:

- RS1: Community Capacity Development (6)
- GV6: Democratization of External Processes (2)
- GV1: Decentralized Authority Distribution (2)
- PO1: Policy and Systemic Transformation Advocacy (2)

- SC1: Inclusive Participation Structures (2)

Case Count: 6 cases

Cases: DAIR (Distributed AI Research Institute), First Nations Information Governance Centre (FNIGC), Masakhane, Participatory Brazil, Slum Dwellers International, Stop LAPD Spying Coalition

Conditions combine as follows to produce this outcome: RS1 (Community Capacity Development) *builds capacity* through direct investment in community skills, knowledge, and organizational development, while GV6 (Democratization of External Processes) *externalizes building* by creating mechanisms for communities to engage with and transform external institutions. GV1 (Decentralized Authority Distribution) *enables agency* by distributing decision-making authority that communities can then exercise, and PO1 (Policy and Systemic Transformation Advocacy) *advocates capacity* by securing policy support for community capacity building. SC1 (Inclusive Participation Structures) *structures capacity* by ensuring capacity building reaches marginalized groups. Together, these conditions produce capacity building impact through direct investment, authority distribution, and inclusive structures—RS1 is the primary mechanism, GV1 and GV6 create the governance conditions for capacity to be exercised, PO1 secures external support, and SC1 ensures inclusivity.

IM3: Cultural and Linguistic Sovereignty

Group: Impact

Top Conditions Present:

- PU2: Cultural Preservation and Linguistic Sovereignty (3)
- PT3: Community Sovereignty Protection in Partnerships (2)
- PR3: Cultural Erasure and Heritage Threat (2)
- TT2: Culturally-Grounded Technology Artifacts (2)

Case Count: 3 cases

Cases: Masakhane, Monlam AI, Te Hiku Media

Conditions combine as follows to produce this outcome: PU2 (Cultural Preservation and Linguistic Sovereignty) *advances sovereignty* through explicit purpose orientation toward cultural and linguistic self-determination, while PT3 (Community Sovereignty Protection in Partnerships) *guards sovereignty* by ensuring external partnerships don't compromise cultural integrity. PR3 (Cultural Erasure and Heritage Threat) *counters threat* by motivating action against ongoing cultural erasure, and TT2 (Culturally-Grounded Technology Artifacts) *preserves culture* by creating technology that embodies and transmits cultural knowledge. Together, these conditions produce cultural sovereignty impact through purpose-driven preservation, protective partnerships, and culturally-embedded technology—PU2 provides direction, PT3 protects external relationships, PR3 provides urgency, and TT2 creates preservation mechanisms.

IM4: Economic Value Transformation

Group: Impact

Top Conditions Present:

- RS3: Economic Value Redistribution (3)

Case Count: 3 cases

Cases: Guifi.net, Namma Yatri, The Drivers Cooperative

Conditions combine as follows to produce this outcome: RS3 (Economic Value Redistribution) *transforms flows* by restructuring how economic value is distributed, shifting from extractive to redistributive patterns. This condition operates as the primary mechanism for economic transformation, working through ownership structures, pricing mechanisms, and value retention strategies that redirect economic flows toward workers, communities, and local contexts rather than external shareholders or platforms.

IM5: Governance and Institutional Transformation

Group: Impact

Top Conditions Present:

- RS1: Community Capacity Development (3)
- TP1: Democratic Data Authority (2)
- GV4: Democratic Scaling Strategies (2)
- GV6: Democratization of External Processes (2)
- PT4: Expertise-Community Knowledge Integration (2)
- PO1: Policy and Systemic Transformation Advocacy (2)

Case Count: 6 cases

Cases: CLEAR (Civic Laboratory for Environmental Action Research), CoopCycle, First Nations Information Governance Centre (FNIGC), OpenSAFELY, Participatory Brazil, Slum Dwellers International

Conditions combine as follows to produce this outcome: RS1 (Community Capacity Development) *enables transformation* by building the community capacity needed to engage with and reshape institutions, while TP1 (Democratic Data Authority) *democratizes governance* by creating data governance mechanisms that transform how institutions relate to data subjects. GV4 (Democratic Scaling Strategies) *scales governance* by creating governance innovations that can expand beyond initial contexts, and GV6 (Democratization of External Processes) *transforms institutions* by applying democratic practices to external institutional engagement. PT4 (Expertise-Community Knowledge Integration) *translates*

expertise by bridging technical and community knowledge for institutional engagement, and PO1 (Policy and Systemic Transformation Advocacy) *advocates transformation* by engaging policy arenas to demand institutional change. Together, these conditions produce governance transformation through capacity building, data democratization, scaling, external engagement, and advocacy—RS1 and PT4 build the internal capacity, TP1, GV4, and GV6 create governance innovations, and PO1 engages external systems.

IM6: Climate and Ecological Transformation

Group: Impact

Top Conditions Present:

- SC3: Climate Crisis and Ecological Response (2)

Case Count: 3 cases

Cases: Slum Dwellers International, SOLshare, Som Energia

Conditions combine as follows to produce this outcome: SC3 (Climate Crisis and Ecological Response) *responds to crisis* by grounding initiatives in explicit ecological purpose and climate adaptation. This condition operates as the primary orientation for climate transformation, working through initiatives that directly address climate vulnerability, energy transition, and ecological resilience through community-led approaches.